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Invited Review

Educational timetabling: Problems, benchmarks, and state-of-the-art results

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ABSTRACT

We propose a survey of the research contributions on the field of Educational Timetabling with a specific focus on "standard" formulations and the corresponding benchmark instances. We identify six of such formulations and we discuss their features, pointing out their relevance and usability. Other available formulations and datasets are also reviewed and briefly discussed. Subsequently, we report the main state-of-the-art results on the selected benchmarks, in terms of solution quality (upper and lower bounds), search techniques, running times, and other side settings.

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1. Introduction

Educational Timetabling, in essence, consists in assigning teacher/student meetings to days, timeslots, and classrooms. Despite this apparent simplicity, experience teaches us that every single institution has its own rules, conventions, and fixations, thus making each specific problem almost unique. As a consequence, many different problem formulations have been proposed in the literature on Educational Timetabling, depending on the type of institution (high-school, university, or other), the type of meetings (lectures, exams,...), and the different settings, constraints, and objectives.

Many papers in the literature tackle a specific problem using a selected search method. The authors normally claim the success of the application, though rarely dispelling the doubt over the readers that the method used was more the authors' "favorite" rather than the most suitable for the problem under consideration. A few previous surveys have tried to put in order this situation by creating a taxonomy of both problem formulations and corresponding search methods used for their solution, in order to draw some conclusions about what works best in each specific case (see Section 2).

In this survey, we want to take a somewhat different point of view. Specifically, we focus on the review of the problem formulations and their publicly available datasets, critically discussing their practical relevance and usability. To this aim, we highlight which datasets have been considered most frequently in the literature, so that they have risen to the status of *benchmarks*, and the corresponding formulation to the status of a *de facto* standard.

We identified six standard formulations, which are presented in chronological order in Sections 3.1–3.6. Remarkably, the chronological order corresponds also to the order of increasing complexity and adherence to the real-world situation. Indeed, we can see that the research has moved continuously from very simplified problems toward full-fledged ones. Nonetheless, in our opinion the early simplified formulations are still interesting testbeds for new search methods, and they have not yet finished to serve their purpose. The main reason is that the accumulated bulk of results and techniques make them a good ground for rigorous comparisons and analyses, given also that most instances have not been solved to optimality yet.

For these formulations and benchmarks, we discuss state-of-the-art results, in terms of solution quality, search techniques, running times, and other side settings. Then we will discuss the availability of upper and lower bounds, in order to identify which are the most challenging instances for future comparisons.

We also review and discuss other formulations that have not attracted general interest so far, but still provide real-world publicly available datasets and could be potentially interesting for the community.

Finally, we consider the issue of reliability of the results claimed in the literature, stressing the importance of the presence of instance and solution checkers, so as to provide against possible errors and misunderstandings. To this aim, we developed a web application, named OptHub (https://opthub.uniud.it), that allows users to check and upload both new instances and solutions. All data, properly validated and timestamped, is available for down-

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load and inspection, along with scoreboards and statistics. The system is meant to provide a unified and up-to-date site for current contributions, so as to facilitate and encourage further research and future comparisons. OptHub, whose development is still ongoing, currently hosts four of the formulations discussed in this survey. The formulations hosted are the early ones that do not have a dedicated and updated online repository on their own.

In a way, this survey is meant for researchers interested in writing what Johnson (2002) called a *horse race paper*, in which the authors assess the quality of their methods by the comparison to previous research on the designated benchmarks. We aim to help such perspective researchers to be rigorous, fair, and comprehensive as much as possible in such a complex task of comparing with the whole literature.

However, our hope is that this effort could be useful also for the authors of an *application paper* (still following Johnson's terminology), that aims at solving one specific original problem. Indeed, those authors could evaluate the quality of their search method by identifying an underlying standard problem that could be a simplified version of their own specific one, adapt their search method to solve it, and report the corresponding results. Naturally, it is not expected that a solver for a complex, full-fledged problem could outperform specialized ones for the benchmarks, but this would give a reasonable measure of the quality of the proposed approach.

This survey is organized as follows. In Section 2, we list the various problems within the scope of the Educational Timetabling area. In Section 3, we introduce and discuss the available formulations and datasets for these problems. We illustrate and comment the state-of-the-art results for the benchmarks in Section 4. Finally, conclusions and future directions are discussed in Section 5.

Web links reported in this survey are not included in the bibliography in order to keep it cleaner. All of them have been visited on July 4th, 2022.

2. Educational timetabling

In this section, we introduce the educational timetabling problems and discuss various general issues of the research area.

2.1. Educational timetabling problems

According to the literature on timetabling (see, e.g., Kingston, 2013; Schaerf, 1999), there are three main problems in the educational timetabling area:

- **High-School Timetabling (HTT)** The weekly scheduling for all the classes of a high-school, avoiding teachers meeting two classes at the same time, and vice versa.
- **University Course Timetabling (CTT)** The weekly scheduling for all the lectures of a set of university courses, avoiding as much as possible the overlap of lectures of courses having common students.
- **University Examination Timetabling (ETT)** The scheduling for the exams of a set of university courses, avoiding overlap of exams of courses having common students, and spreading the exams for the students as much as possible.

Even though a clear cut between HTT, CTT, and ETT is not possible (e.g., some high-schools are organized in a university fashion), they normally differ from each other significantly, and most of the papers in the literature can be classified within one of these three problems.

2.2. Previous surveys

Many surveys on educational timetabling have appeared in the literature. However, due to the vastness of the research area, all of them focus an a subset of the problems introduced in the previous section, in order to reduce their scope. For example, Burke & Petrovic (2002) and MirHassani & Habibi (2013) focus on university timetabling (CTT and ETT), likewise Lewis (2008) who further limits his study to metaheuristic techniques. Similarly, the survey by Qu, Burke, McCollum, Merlot, & Lee (2009) is dedicated to ETT, whereas the one by Pillay (2014) is only on HTT, and the recent ones by Chen, Sze, Goh, Sabar, & Kendall (2021) and Tan, Goh, Kendall, & Sabar (2021) are on CTT and HTT, respectively. Lastly, the survey by Bettinelli, Cacchiani, Roberti, & Toth (2015) reviews only one specific formulation of CTT, namely the curriculum-based course timetabling (CB-CTT), that will be introduced and discussed in Section 3.3.

2.3. Other timetabling problems

There are also other problems within the Educational Timetabling field that have been addressed in the literature, although they are less popular than the previous three. Among these "minor" problems we can include *Student Sectioning* (Müller & Murray, 2010), *Thesis Defense Timetabling* (Battistutta, Ceschia, De Cesco, Di Gaspero, & Schaerf, 2019), *Trainee/Intern/Resident Assignment* (for medical and military schools) (Akbarzadeh & Maenhout, 2021), and *Conference Scheduling* (Stidsen, Pisinger, & Vigo, 2018). We do not discuss the above problems in details, as there are no available datasets that have reached the status of benchmarks. An exception is Student Sectioning that is included together with CTT in the ITC-2019 formulation, that will be discussed in Section 3.6.

Other timetabling problems, which fall outside the scope of Educational Timetabling, such as Employee Timetabling (Meisels & Schaerf, 2003), Transportation (trains and airplanes) Timetabling (Cacchiani & Toth, 2012), and Sports Timetabling (Van Bulck, Goossens, Schönberger, & Guajardo, 2020) are not discussed here.

2.4. Timetabling initiatives

The timetabling community is quite active. There are a biannual conference series (http://patatconference.org) and a EURO Working Group (https://www.euro-online.org/web/ewg/14/), both called PATAT (Practice and Theory of Automated Timetabling) and dedicated to the whole area of Timetabling problems. One of their activities has been the organization of a range of competitions, and in particular five International Timetabling Competitions: ITC-2002, ITC-2007 (McCollum et al., 2010), ITC-2011 (Post, Di Gaspero, Kingston, McCollum, & Schaerf, 2016), ITC-2019 (Müller, Rudová, & Müllerová, 2018), and ITC-2021 (Van Bulck, Goossens, Belien, & Davari, 2021). These competitions have brought forth most of the standard formulations and benchmarks discussed in Section 3. Incidentally, the most recent one, ITC-2021, did not focus on educational timetabling like the previous ones but on sports timetabling. The other competitions organized by PATAT regard the Nurse Rostering problem: the First International Nurse Rostering Competition (INRC-I) (Haspeslagh, De Causmaecker, Schaerf, & Stølevik, 2014) and the Second International Nurse Rostering Competition (INRC-II) (Ceschia, Dang, De Causmaecker, Haspeslagh, & Schaerf, 2019).

2.5. Multiobjective formulations

For all the standard formulations that we will introduce in Section 3, there is a single objective function, defined as a weighted sum of the various penalty terms to be minimized. Therefore, we do not include in this survey the issues related to multiobjective optimization (Silva, Burke, & Petrovic, 2004), although the multiobjective perspective would be surely useful in

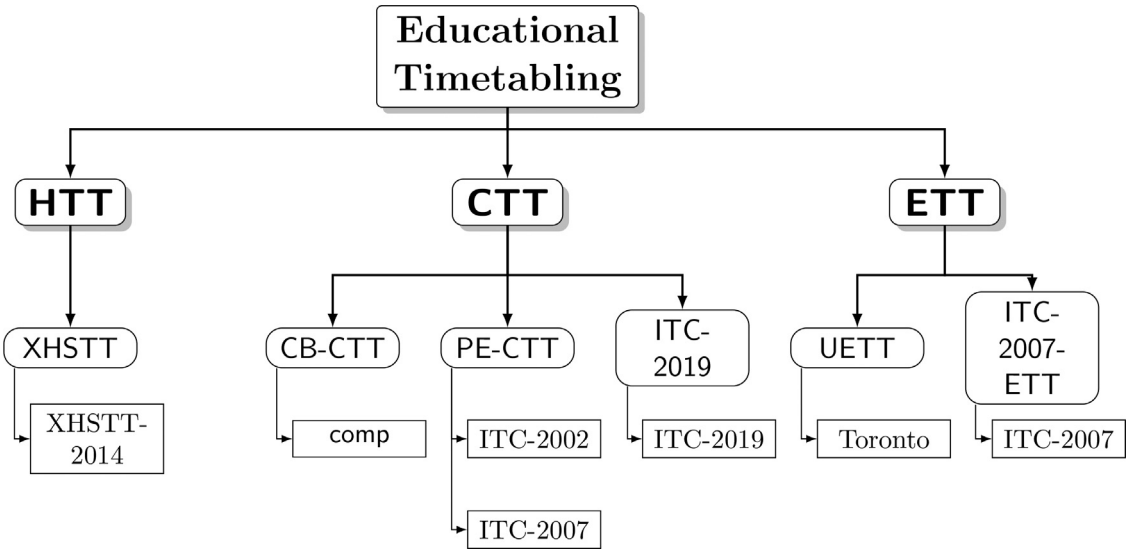


Fig. 1. Educational timetabling problems, formulations, and benchmarks.

this context, as objectives in timetabling could be rather intangible and thus not always commensurable. Indeed, many objectives are related to the comfort of the participants (students or teachers), so that it is difficult to assign to them a specific numeric weight. Furthermore, besides the classical objectives measuring the general comfort, some authors include also other notions, which are even more difficult to be put in the same scale of the other objectives. These include the *fairness* (Mühlenthaler & Wanka, 2016), that takes care for the balanced distribution of the discomfort among the participants (teachers and students) and the *robustness* (Akkan & Gülcü, 2018), that measures the possibility to not deteriorate the quality in presence of unforeseen disruptions of the timetable.

2.6. Terminology and taxonomy

We define here some common terms in the timetabling vocabulary that will be used throughout this survey. Concepts that are specific of one formulation are introduced in the dedicated section.

- **Times:** The time *horizon* is divided into *days* and each day is split into *timeslots* (in general, the same number of timeslots is given in each day). A *period* is a pair ⟨day, timeslot⟩.
- **Events:** An *event* is a meeting between students and one or more teachers. Events can be of different types: *lectures* or *exams* of a *course*, *laboratories*, or *seminars*.
- **Resources:** We consider three main kinds of resources: students, teachers, and rooms. Events have to be scheduled taking into account resource restrictions, such as *students*' enrollments, *teachers*' requests and *rooms*' availabilities.
- **Constraints:** As customary, constraints are split into *hard* and *soft* ones (soft constraints are also called objectives). The hard constraints must always be satisfied, whereas the soft ones contribute to the objective function, which is a weighted sum of all soft constraint penalties.

Figure 1 shows the taxonomy of the problems presented in Section 2.1, and the corresponding formulations introduced one by one in Section 3. For each formulation, the figure reports the datasets used as benchmarks.

3. Formulations and datasets

We introduce the selected formulations and the corresponding datasets in Sections 3.1–3.6. For each of these formulations, we present in turn (*i*) a brief specification, (*ii*) the benchmarks with their main features, (*iii*) the file formats and their usability, (*iv*) the presence of additional datasets and instance generators, and (*v*) some discussion, in particular the assessment of the gap w.r.t. the complete real-world problem.

From the benchmarks, we identify and remove the instances that are too easy to be kept in the pool, and their presence has little value in challenging and comparing different search algorithms. We name an instance as easy when all runs of the top search techniques always find the optimal value.

Finally, Section 3.7 is devoted to list and discuss the other real-world formulations that provide available (and usable) public datasets.

3.1. Uncapacitated examination timetabling (UETT)

The first formulation that we consider is the classical version of ETT proposed by Carter, Laporte, & Lee (1996), that we name UETT (U for uncapacitated, as explained below). This is a very essential view of the examination timetabling problem, which extends just slightly the underlying *graph coloring* problem, with exams as nodes and periods as colors.

Short specification The main input data of UETT is the Boolean-valued *enrollment* matrix, that stores for each pair ⟨student, exam⟩ the information about whether the student has to take the exam or not.

Two exams with at least one student in common are in conflict, so that they cannot to be scheduled in the same period. Conflicts are the sole constraints. In particular, rooms are not taken into account, and for this reason the problem is known as *uncapacitated*.

The objective function is related to the distance between exams with students in common. Distances are penalized in the following fixed way: the cost of scheduling two exams with k students in common at distance of 1, 2, 3, 4, and 5 periods is $16k$, $8k$, $4k$, $2k$, and k, respectively.

Benchmarks The UETT formulation comes with a dataset of 13 real-world instances mainly from North American universities, known as Toronto instances (or Carter's instances), whose main features are illustrated in Table 1 (adapted from Bellio, Ceschia, Di Gaspero, & Schaerf, 2021).

As remarked by Alefragis, Gogos, Valouxis, & Housos (2021), the instances have some unnecessary data that could be removed by preprocessing. Indeed, some students are enrolled in only one exam, so that they can affect neither the constraints nor the objec-

Table 1
Features of the Toronto benchmark instances. Symbol definition: **E** (Exams), **S** (Students), **P** (Periods), **ES** (Exams per Student), **CGD** (Conflict Graph Density).

Inst.	E		S		P	ES		CGD
	total	active	total	active		avg	max	
car91	682	678	16,925	13,516	35	4.20	9	0.13
car92	543	542	18,419	14,450	32	3.84	7	0.14
ear83	190	190	1125	1124	24	7.21	10	0.27
hec92	81	81	2823	2502	18	4.25	7	0.42
kfu93	461	444	5349	5073	20	4.92	8	0.06
lse91	381	379	2726	2627	18	4.16	8	0.06
pur93	2627	2413	30,032	27,405	42	4.40	9	0.03
rye93	486	485	11,483	9458	23	4.76	10	0.08
sta83	139	139	611	611	13	9.41	11	0.14
tre92	261	260	4360	3693	23	4.03	6	0.18
uta92	622	622	21,266	15,086	35	3.91	7	0.13
ute92	184	184	2750	2672	10	4.41	6	0.08
yor83	181	181	941	940	21	6.42	14	0.29

tive. These students are called *noise students* by Alefragis et al. In turn, exams taken only by noise students do not contribute to the constraints and the objective, and they are called *noise exams*. In Table 1, we report both the total number of students and exams and the *active* (non-noise) ones. The two columns **ES** represent the number of exams per active student (average and maximum value). The rightmost column **CGD** is the density of the conflict graph, which is computed as the number of conflicts divided by $n \cdot (n-1)/2$, where n is the number of active exams.

File formats and repositories. Instances are available in plain text and split in two separate files: one containing the exams and one with the student enrollments. The files were originally posted via FTP in the website of the University of Toronto (not active anymore), and are now available at http://www.cs.nott.ac.uk/~pszrq/data.htm. The same instances are posted on OptHub with a slightly modified (more robust) single-file format.

The original data format is unfortunately very fragile, as for example the accidental insertion of a newline character would result in a different (but still valid) instance. This has actually happened as discussed below.

Other datasets and generators. Other instances of UETT are available. First, there is a set of 9 instances, called *apocryphal* by Bellio et al. (2021), that are variants of some of Toronto ones that were created by accidental perturbation of the original files and used unwittingly in a few experimental analyses (see Qu et al., 2009, for a discussion about them). Even though they have been considered by a few authors, given that they are just arbitrary perturbations of the real instances, we do not consider them as benchmarks.

Another set of 20 instances, obtained by translating real-world instances for other examination timetabling formulations, have been made available by Bellio et al. (2021) on OptHub. Finally, Bellio et al. developed a parametric generator that creates artificial instances with the prescribed values of the main features. A set of 100 generated instances, selected based on feasibility and computational hardness, are also available on OptHub.

Discussion. UETT is surely a simplified formulation, as the authors themselves admit that "all side constraints have been removed" (Carter et al., 1996). Indeed, they list in their original work a set of constraints that apply to some of the real-world cases, but have been neglected in the proposed formulation, in order to have a common ground for many different cases.

Despite its extreme simplicity, or perhaps actually due to it, UETT has been and still is an active subject of studies (see Section 4.1). The main reason could also be that the benchmarks proposed are very challenging. In fact, to the best of our knowledge, only one of such instances has been solved to proven optimality so far.

3.2. Post-Enrolment course timetabling (PE-CTT)

The second formulation that we consider is the so-called Post-Enrolment Course Timetabling (PE-CTT) problem that is the first standard formulation of CTT. It has been proposed within the Metaheuristics Network project (2000–04), then used as the subject of ITC-2002, and used again for ITC-2007 with a slightly more complex formulation, which is the one discussed here. The full specification can be found in the work by Lewis, Paechter, & McCollum (2007, §3).

Short specification. In PE-CTT a set of events, a set of periods, and a set of rooms are given. Also a set of days is defined, and each period is a timeslot belonging to one day. Students enroll in events causing conflicts between them. Conflicting events cannot be scheduled in the same period.

Furthermore, there is a set of room features that may be required by events. Room features and capacity (in terms of seats) together result in a compatibility relation between rooms and events.

In addition, precedence relations between events are defined, that impose that some events must be scheduled before others. The last constraints are the ones originated from an unavailability relation, stating that an event cannot be scheduled in some specified periods.

The objective function is composed by three components that penalize the following cases: (*i*) a student attending an event in the last timeslot of a day, (*ii*) a student attending three (or more) events in successive timeslots in the same day, (*iii*) a student attending only one event in a day.

Benchmarks. There are two datasets that can be considered as consolidated benchmarks for PE-CTT, which are the ones coming from the competitions ITC-2002 and ITC-2007. The dataset from ITC-2002 is on a simplified version of the problem that does not consider precedences and unavailabilities.

The main features of the instances are illustrated in Tables 2 and 3. The conflict graph density (**CGD**) is computed as in Section 3.1 with events in place of exams. The room occupancy (**RO**) is the ratio between events and rooms per periods (does not consider the capacity). The number of periods is 45 for all instances, thus it is not listed in the tables. All instances are artificial and obtained by a generator. In addition, they have the peculiarity of having been generated in such a way that at least one *perfect* (zero cost) solution exists.

For ITC-2007 instances, it is generally more difficult than ITC-2002 ones to find a feasible solution. Indeed, looking at the tables, we see that they have a higher conflict graph density (column **CGD**), in addition to unavailabilities and precedences that are absent in the ITC-2002 version.

Table 2
Features of the ITC-2002 benchmark instances. Symbol definition: **E** (Events), **R** (Rooms), **S** (Students), **RO** (Room Occupancy), **CGD** (Conflict Graph Density), **SE** (Students per Event), **ES** (Events per Student), **RE** (suitable Rooms per Event).

Inst.	E	R	S	RO	CGD	SE	ES	RE
01	400	10	200	0.89	0.20	8.88	17.75	1.96
02	400	10	200	0.89	0.21	8.61	17.23	1.92
03	400	10	200	0.89	0.23	8.85	17.70	3.42
04	400	10	300	0.89	0.23	13.07	17.43	2.45
05	350	10	300	0.78	0.31	15.24	17.78	1.78
06	350	10	300	0.78	0.26	15.23	17.77	3.59
07	350	10	350	0.78	0.21	17.48	17.48	2.87
08	400	10	250	0.89	0.17	10.99	17.58	2.93
09	440	11	220	0.89	0.17	8.68	17.36	2.58
10	400	10	200	0.89	0.20	8.89	17.78	3.49
11	400	10	220	0.89	0.20	9.58	17.41	2.07
12	400	10	200	0.89	0.20	8.79	17.58	1.96
13	400	10	250	0.89	0.21	11.06	17.69	2.43
14	350	10	350	0.78	0.25	17.42	17.42	3.08
15	350	10	300	0.78	0.25	15.07	17.58	2.19
16	440	11	220	0.89	0.18	8.88	17.75	3.17
17	350	10	300	0.78	0.31	15.15	17.67	1.11
18	400	10	200	0.89	0.21	8.78	17.56	1.75
19	400	10	300	0.89	0.20	13.28	17.71	3.94
20	350	10	300	0.78	0.25	14.99	17.49	3.43

Table 3
Features of ITC-2007 benchmark instances. Symbol definition: **E** (Events), **R** (Rooms), **S** (Students), **RO** (Room Occupancy), **CGD** (Conflict Graph Density), **SE** (Students per Event), **ES** (Events per Student), **RE** (suitable Rooms per Event), **TE** (availability of Timeslots for Event), **PE** (number of Preceding events per Event).

Inst.	E	R	S	RO	CGD	SE	ES	RE	TE	PE
01	400	10	500	0.89	0.34	26.27	21.02	4.08	0.56	0.10
02	400	10	500	0.89	0.37	26.29	21.03	3.95	0.57	0.09
03	200	20	1000	0.22	0.47	66.92	13.38	5.05	0.56	0.10
04	200	20	1000	0.22	0.52	66.98	13.40	6.40	0.57	0.10
05	400	20	300	0.44	0.31	15.69	20.92	6.80	0.56	0.37
06	400	20	300	0.44	0.30	15.55	20.73	5.07	0.56	0.35
07	200	20	500	0.22	0.53	33.67	13.47	1.58	0.39	0.10
08	200	20	500	0.22	0.52	34.58	13.83	1.91	0.38	0.10
09	400	10	500	0.89	0.34	26.78	21.43	2.91	0.56	0.11
10	400	10	500	0.89	0.38	26.23	20.98	3.20	0.56	0.10
11	200	10	1000	0.44	0.50	68.04	13.61	3.38	0.56	0.10
12	200	10	1000	0.44	0.58	68.04	13.61	3.36	0.57	0.10
13	400	20	300	0.44	0.32	15.90	21.19	8.68	0.56	0.34
14	400	20	300	0.44	0.32	15.64	20.86	7.56	0.56	0.36
15	200	10	500	0.44	0.54	32.63	13.05	2.23	0.38	0.10
16	200	10	500	0.44	0.46	34.10	13.64	1.74	0.39	0.11
17	100	10	500	0.22	0.71	97.67	19.53	2.77	0.57	0.12
18	200	10	500	0.44	0.65	51.42	20.57	3.47	0.57	0.10
19	300	10	1000	0.67	0.47	44.78	13.44	3.66	0.56	0.10
20	400	10	1000	0.89	0.28	33.92	13.57	3.73	0.56	0.10
21	500	20	300	0.56	0.23	12.40	20.67	7.36	0.57	0.36
22	600	20	500	0.67	0.26	17.42	20.90	5.65	0.56	0.39
23	400	20	1000	0.44	0.44	53.42	21.37	2.89	0.78	0.12
24	400	20	1000	0.44	0.31	33.34	13.34	1.59	0.55	0.72

File formats and repositories. All instances are available in a lengthy text-only format, in which all elements of the matrices are written explicitly, one per line. As a consequence, the files are easy to parse, but rather verbose, not human readable, and fragile. They are available at the websites of the competitions, reachable from the PATAT conference website (http://patatconference.org/). Instances are available also on OptHub.

Other datasets and generators. Two other datasets are publicly available, and they have been considered in some papers, though less frequently than the two mentioned above. The first one is a dataset proposed by the Metaheuristics Network (see Rossi-Doria et al., 2003) before the competitions, using the simplified formulation of ITC-2002, which are available at http://iridia.ulb.ac.be/supp/IridiaSupp2002-001. For recent results on these instances see for example (Goh, Kendall, & Sabar, 2017).

The second dataset, including much larger instances, has been introduced by Lewis & Paechter (2007) with the aim of having more difficult cases and is available at http://www.rhydlewis.eu/hardTT/. Indeed, for these instances, feasibility is quite difficult to be obtained, and the comparison is on the number of violations rather than on the objective function. For results on these instances see for example (Ceschia, Di Gaspero, & Schaerf, 2012).

The generator used for the instances of ITC-2002 and ITC-2007 was never made public, and no other one has been developed and made available for this formulation.

Discussion. Like UETT discussed in the previous section, PE-CTT is rather a simplified formulation with respect to the real-world problem, as many aspects are deliberately removed in order to make the problem more manageable (see Lewis et al., 2007, §5 for a discussion about the neglected features).

In addition, as shown in Tables 2 and 3, the diversity of the features of the instances is quite limited. For example, all instances have exactly 45 periods divided in 5 days of 9 timeslots each, with no variability at all. Similarly, the number of rooms is restricted to just three different values, namely 10, 11, and 20.

3.3. Curriculum-Based course timetabling (CB-CTT)

The next formulation is the so-called Curriculum-based Course Timetabling (CB-CTT), which has been proposed by Di Gaspero & Schaerf (2003), and subsequently adopted, in a slightly modified version, as the third track of ITC-2007.

The name of the problem comes from the fact that conflicts are determined by predefined curricula, opposed to the use of explicit student enrolments as in PE-CTT. This is however not the most important difference between PE-CTT and CB-CTT as the notions of student and curriculum are formally interchangeable, because a student can be expressed as a curriculum and vice versa. On the contrary, the main difference stems from the notion of *course* as a set of lectures that is absent in PE-CTT. Many constraints and objectives in CB-CTT are defined at the level of a course, whereas in PE-CTT constraints and objectives are always expressed at the level of the single event/lecture.

A few variants of this formulation have been subsequently proposed by Bonutti, De Cesco, Di Gaspero, & Schaerf (2012). The most studied one however remains the one used for ITC-2007 (named UD2 in Bonutti et al., 2012), which is thus the one that we consider here. The full description is provided by Di Gaspero, McCollum, & Schaerf (2007).

Short specification. As mentioned above, the key notions of CB-CTT are courses and curricula. Each course consists of a fixed number of *lectures* to be scheduled in different periods. A course is attended by a number of *students*, and is taught by a *teacher*. For each course, there are a minimum number of days over which the lectures of the course should be spread. Moreover, there are some unavailable periods in which the course cannot be scheduled.

Like in PE-CTT, we are given a number of periods divided in days and timeslots in the day. Each *room* has a *capacity*, specified as the number of available seats, but no other features.

A *curriculum* is a group of courses that potentially have students in common. As a consequence, lectures of courses belonging to the same curriculum are in conflict and cannot be scheduled in the same period. Two courses are in conflict also if they are taught by the same teacher.

The hard constraints regard conflicts, teacher availability and room occupancy. The objective function (soft constraints) is composed by four components that penalize the following cases: (*i*) the capacity of the room assigned to a lecture is less than the number of students attending the course, (*ii*) the lectures of a course are not spread into the given minimum number of days, (*iii*) a lecture is isolated, i.e., not adjacent to any other in the same curriculum, (*iv*) the lectures of a course are not given all in the same room.

Benchmarks. A few real-world datasets are available for the formulation. By far the most used one is the one from ITC-2007, known as the comp dataset that we consider as benchmark. This dataset is composed by real-world instances coming from University of Udine (Italy), mainly from the School of Engineering.

Table 4 (taken from Bonutti et al., 2012) shows the main features of these instances. It could be noticed that we removed instances comp11 and comp15. They have been excluded for different reasons: comp11 is an easy one, as all competitive search methods always find a solution of cost zero, while comp15 is actually identical to comp03 in the problem variant UD2 that we consider here.

File formats and repositories. Instances are available in an ad-hoc text-only format, which is reasonably human-readable. There are actually two versions of the format, the original .ctt one used for the competition, and the newer .ectt (e for extended) proposed by Bonutti et al. (2012) that includes additional data necessary for the other versions of the problem.

Instances are available on OptHub in .ectt format along with several results from the literature. The solutions were imported from the original CB-CTT website (satt.diegm.uniud.it/ctt) not available anymore.

Other datasets and generators. A few other datasets of real-world instances coming mainly from Italian universities are available on OptHub.

An instance generator has been developed by Burke, Mareček, Parkes, & Rudová (2008), which has been subsequently revised by Lopes & Smith-Miles (2010, 2013) in such a way to obtain more realistic instance, in particular more similar to the comp dataset. The latter generator has been further refined by De Coster, Musliu, Schaerf, Schoisswohl, & Smith-Miles (2022) in order to enlarge the region of the instance space covered by the generated instances. The generator by De Coster et al. is available at https://cdlab-artis.dbai.tuwien.ac.at/papers/cb-ctt/.

Discussion. Like UETT and PE-CTT, the CB-CTT formulation is a judicious simplification of the original problem. Constraints and objectives included in the formulation have been selected among the long list of real ones to be general and simple enough, but also representative of the various types of restrictions. For example, the objective on room stability for the lectures of a course, which is meant to improve the comfort of teachers and students, is not particularly important in practice, but it represents a set of limitations that involve the use of rooms in different periods. Without this objective the management of the rooms could have been done independently for each period, which would have resulted in an over-simplification of the problem.

The comp instances are extracted from various departments of University of Udine (Italy), so that the values of the main features are quite diverse. The additional instances come also from different universities, so that they are yet broader in size and structure.

3.4. Examination timetabling (ITC-2007-ETT)

The next formulation that we include in our study is the ETT proposed for ITC-2007 (Track 1). This formulation, even though it does not consider all practical features, is much more realistic than the uncapacitated version discussed in Section 3.1. Indeed, it includes several novel features collected from the activity of a commercial software in use in many British universities. The full specification is provided by McCollum, McMullan, Burke, Parkes, & Qu (2007).

Short specification. Like other formulations, the time horizon is divided in a number of periods, each one belonging to a day. The novelty is that periods have a specific length (in minutes) and can have a penalty for scheduling exams in it.

As usual, rooms have a capacity and might be undesired, in the sense that there is a penalty for their use, like periods.

For each exam, a length of execution is given, so that it is compatible only with periods of sufficient duration. In addition, an exam might require to be scheduled in a dedicated room, otherwise it can share the room with other exams. For each exam, the set of students enrolled is given.

For some pairs of exams a precedence rule is specified, stating that one exam must be scheduled after, at the same time, or at a different time with respect to the other one.

The objective function is composed by the following components (soft constraints): (*i*) a student taking two exams in consecutive periods in the same day, (*ii*) a student taking two exams in

Table 4
Features of comp benchmark instances. Symbol definition: **C** (Courses), **L** (Lectures), **R** (Rooms), **PD** (Periods per Day), **D** (Days), **Cu** (Curricula), **MML** (Min and Max Lectures per day per curriculum), **Co** (number of Conflicts per Course), **TA** (Teacher Availability), **CL** (number of Lectures per Curriculum per day), **RO** (Room Occupancy).

Inst.	C	L	R	PD	D	Cu	MML	Co	TA	CL	RO
comp01	30	160	6	6	5	14	2–5	13.2	0.93	3.24	0.88
comp02	82	283	16	5	5	70	2–4	7.97	0.76	2.62	0.70
comp03	72	251	16	5	5	68	2–4	8.17	0.78	2.36	0.62
comp04	79	286	18	5	5	57	2–4	5.42	0.81	2.05	0.63
comp05	54	152	9	6	6	139	2–4	21.7	0.59	1.8	0.46
comp06	108	361	18	5	5	70	2–4	5.24	0.78	2.42	0.80
comp07	131	434	20	5	5	77	2–4	4.48	0.80	2.51	0.86
comp08	86	324	18	5	5	61	2–4	4.52	0.81	2	0.72
comp09	76	279	18	5	5	75	2–4	6.64	0.81	2.11	0.62
comp10	115	370	18	5	5	67	2–4	5.3	0.77	2.54	0.82
comp12	88	218	11	6	6	150	2–4	13.9	0.57	1.74	0.55
comp13	82	308	19	5	5	66	2–3	5.16	0.79	2.01	0.64
comp14	85	275	17	5	5	60	2–4	6.87	0.75	2.34	0.64
comp16	108	366	20	5	5	71	2–4	5.12	0.81	2.39	0.73
comp17	99	339	17	5	5	70	2–4	5.49	0.79	2.33	0.79
comp18	47	138	9	6	6	52	2–3	13.3	0.64	1.53	0.42
comp19	74	277	16	5	5	66	2–4	7.45	0.76	2.42	0.69
comp20	121	390	19	5	5	78	2–4	5.06	0.78	2.5	0.82
comp21	94	327	18	5	5	78	2–4	6.09	0.82	2.25	0.72

Table 5
Features of the ITC-2007 benchmark instances. Symbol definition: **E** (Exams), **S** (Students), **P** (Periods), **R** (Rooms), $\mathbf{P}_{HC}$ (number of hard constraints on periods), $\mathbf{R}_{HC}$ (number of hard constraints on rooms), **PF** (Periods involved in the Frontload constraint), **EF** (Exams involved in the Frontload constraint), **PS** (Periods Spread: minimum required distance among exams per student), **CGD** (Conflict Graph Density), **ER** (ratio between Exams and Rooms per periods), **SE** (Students per Exam), **S/Cap** (total Students in exams divided by total Capacity for all periods). The values highlighted in boldface are incoherent with the number of available periods.

Inst.	E	S	P	R	$\mathbf{P}_{HC}$	$\mathbf{R}_{HC}$	PF	/	EF	PS	CGD	ER	SE	S/Cap
1	607	7883	54	7	12	0	30	/	100	5	0.05	1.61	53.3	0.75
2	870	12,484	40	49	8	2	30	/	250	1	0.01	0.44	43.0	0.23
3	934	16,365	36	48	82	15	20	/	200	4	0.03	0.54	65.5	0.33
4	273	4421	21	1	20	0	10	/	50	2	0.15	13.00	79.6	0.86
5	1018	8719	42	3	27	0	30	/	250	5	0.01	8.08	33.6	0.34
6	242	7909	16	8	22	0	**30**	/	25	**20**	0.06	1.9	76.31	0.56
7	1096	13,795	80	15	28	0	30	/	250	10	0.02	0.91	41.5	0.22
8	598	7718	80	8	20	1	**100**	/	250	15	0.05	0.93	52.5	0.43
9	169	624	25	3	10	0	10	/	100	5	0.08	2.25	15.0	0.60
10	214	1415	32	48	58	0	10	/	100	20	0.05	0.14	36.7	0.13
11	934	16,365	26	40	83	15	20	/	400	4	0.03	0.90	65.5	0.48
12	78	1653	12	50	9	7	5	/	25	5	0.18	0.13	47.2	0.20

the same day (*iii*) a student taking two exams within a fixed number of periods (spread), (*iv*) exams in the same room with mixed durations, (*v*) an exam with many students scheduled towards the end of the planning horizon, (*vi*) an exam scheduled in an undesired period or an undesired room.

The weights of the soft constraints vary from case to case, and are included in the input file of each instance.

Benchmarks. A dataset composed of 12 instances from British universities was released for ITC-2007. The main features of these instances[1] are summarized in Table 5 (adapted from Battistutta, Schaerf, & Urli, 2017).

File formats and repositories. The file format is a single-file text-only one, created ad-hoc for the ITC-2007 competition. Although the format is better engineered than the simple one of previous competitions, still there are some fragilities, as it is witnessed by the presence of incoherent data in two of the competition instances.[2] These inconsistencies make the cost of two specific soft constraint types identical in all solutions. Fortunately, this fact does not compromise the significance of those two instances, as all other constraint types already make them sufficiently challenging.

The files are available from the ITC-2007 website and also from OptHub, where there are also a few solutions listed.

Other datasets and generators. A set of 8 instances coming from Yeditepe University and proposed by Özcan & Ersoy (2005) have been translated from their original format to the ITC-2007 one by Parkes & Özcan (2010). They are available at the timetabling web page of Andrew Parkes: http://www.cs.nott.ac.uk/~pszajp/timetabling/exam/. These instance are relatively small, with a maximum size of 210 exams. In addition, they are obtained from a simpler formulation, so that many features of the ITC-2007-ETT formulation are unused. Up to our knowledge, no other real-world instances have been introduced later on for the problem.

An instance generator has been developed by Battistutta et al. (2017) for tuning purposes, and a set of 50 challenging artificial instances has been made public on OptHub.

The original solution checker provided for ITC-2007 is not available anymore, but solutions can be validated from OptHub.

Discussion. As mentioned by McCollum et al. (2007), this formulation is a significant step forward in the use of complete formulations for standard problems. Indeed, with respect to its predecessors it includes many novel real-world features, in particular of British universities, even though, for the aim of simplicity, some aspects are still left out.

This is also the first formulation of our list that has the weights of the different objectives written in the instance, rather than fixed for all scenarios. As written by McCollum et al. (2007): “this is motivated by our experience that different institutions do indeed have

[1] Note that the number of exams reported by Burke & Bykov (2016, Table 2) is overestimated as they consider the largest student identifier, but some numbers are missing in the file.

[2] Instances 6 and 8 have a number of periods involved in the Frontload constraints (both instances) and Period Spread constraint (instance 6 only) larger than the number of periods (highlighted in boldface). This implies that the corresponding soft constraints are always violated, independently of the solution.

different weights, and so no one set would be completely useful". Still from McCollum et al. (2007): "We hope that this will encourage the development of solvers that are robust rather than potentially over-tuned to one particular set of weights for a dataset."

3.5. High-School timetabling (XHSTT)

Our next formulation was introduced by Post et al. (2012) as an attempt to create a unified formulation and data format for the HTT problem. The proposed formulation, called XHSTT, is extremely rich and the intent is to avoid, differently from the previous formulations, any concession to judicious simplifications. In fact, the proposing team is composed by researchers from various countries, with the aim of including the features coming from as many different situations as possible all around the world.

XHSTT has also been used as the subject of the ITC-2011 competition, which has led to a boost for its spread in the scientific community. In fact, XHSTT is the most popular HTT formulation compared to other ones among the community, and it has drawn the attention of many researchers, in particular after ITC-2011. In addition, the dataset is diverse and quite challenging, also compared to previous ones.

Over the years, several versions of the archive have been collected in the XHSTT project, each one mainly based on the previous one with some improvements on the current instances (name change, format simplification, error correction, redundancy removal,...) and some new instances. As a consequence, in some cases authors have competed on slightly different versions of the same instances, so that a comprehensive and fair comparison has not been made possible. Indeed, Kristiansen, Sørensen, & Stidsen (2015) wrote: "such updates to the format make it hard for researchers to compare computational results with those previously reported".

For the full problem specification, we refer to the work by Post et al. (2012) and to the XHSTT website https://www.utwente.nl/hstt/, which contains also some updates with respect to the original formulation.

Short specification. As mentioned above, the formulation is complex, so that it is quite difficult to discuss it in brief. Basically, it includes three types of entities: times, resources (students, classes, teachers, and rooms), and events. For each of these three, it is possible to define sets of atomic elements, and use these sets to express complex constraints and objectives.

In XHSTT there are 15 different types of constraints, which range from spreading lectures in the week, to student idle times, to preferences and unavailabilities. For brevity, we do not list them here and refer again to the XHSTT website for their comprehensive specification. Each individual constraint can be declared either hard (Required) or soft (non-Required).

Benchmarks. (Post et al., 2014) As benchmarks we consider the XHSTT archive (Post et al., 2014), and in particular the current version of the archive at the XHSTT website, called XHSTT-2014. As mentioned in the website: "XHSTT-2014 contains a carefully selected subset of the instances collected during this project, in their most up-to-date form".

This archive is composed of 25 instances. We removed two of them, namely GR-H1-97 and GR-P3-10, for which a perfect solution (i.e., having zero cost) can be easily obtained. The main features of the remaining 23 instances are shown in Table 6 taken from the XHSTT website. The symbol "–" means that the corresponding resource group is omitted in the particular instance.

File formats and repositories. Instances are written in an XML file format, which includes also a metadata part. Thanks to the flexibility of XML, many instances and solutions can be inserted in a single file.

All instances, lower bounds, and best solutions are available at the XHSTT website, including a checker that validates a solution and writes a report of the corresponding violations.

Other datasets and generators. Other instances have been contributed from the community over the years and they have been included in the XHSTT website, but they are currently not interesting. This is because they are considered either too easy to be solved to optimality or too similar to other instances or not completely well-defined. There are also a few artificial ones obtained by translating instances from other formulations, which do not use most of the constraint types. All the instances are available from the XHSTT website. Up to our knowledge, no artificial instance generator is available.

Discussion. As mentioned above, XHSTT is a full-fledged real-world problem with all possible constraints and objectives included. The spirit of this effort is to consider all possible constraints in use somewhere in the globe, allowing the possibility to produce instances that use only a subset of the constraints for its specification. The formulation is still evolving, with student sectioning and different campuses as candidate new features.

A drawback of this choice is that implementing an effective solver for the complete specification of XHSTT is rather labor demanding. However, a solver could also be developed to deal with only a subset of the possible constraint types.

A limit of the benchmark dataset is that nowadays many instances are solved to proven optimality, so that the competition is moved mainly to the performance of solvers under specific timeouts. An alternative standard formulation, which has recently gained some attention, is the Brazilian one introduced by Saviniec & Constantino (2017), mentioned in Section 3.7, and described in the survey by Tan et al. (2021).

3.6. University course timetabling (ITC-2019)

Our last standard formulation is the one of the CTT problem proposed by Müller et al. (2018) for the ITC-2019 competition, that we call ITC-2019. This formulation actually represents a combination of CTT with the student sectioning problem. The formulation is indeed rather rich and structured, and it represents a big step forward bridging the gap between theory and practice of timetabling research. Nonetheless, it still cannot be considered a totally complete problem, as the authors themselves write "to make the problems more attractive, we remove some of the less important aspects of the real-life data while retaining the computational complexity of the problems".

Short specification. ITC-2019 consists in sectioning students into classes based on course enrollments, and then assigning classes to available periods and rooms, respecting various constraints and preferences.

The main novelty is that courses are composed by a set of classes, with one or more configurations, organized in subparts. A student enrolled in a specific configuration of a course must attend exactly one class from each subpart. A class can meet several times a week during certain weeks of the semester. For each class, the list of possible periods and rooms for meetings is given. In addition, each class may have also a parent-child relationship with another class.

The other remarkable feature is that the timetable may differ from week to week, differently from CB-CTT and PE-CTT where the very same weekly timetable is replicated for the whole semester. This feature is present in many practical situations as it allows the institution to gain flexibility in the organization.

In addition, this formulation uses a much finer granularity for the times, setting the length of the timeslot to 5 min for all instances. Conversely, the other CTT formulations set the length of the timeslot typically to 1 h.

Table 6
Features of the XHSTT benchmark instances. Symbol definition: **T** (Times), **Te** (Teachers), **R** (Rooms), **C** (Classes), **S** (Students), **E** (Events), **D** (total Duration of all events).

Inst.	T	Te	R	C	S	E	D
AU-BG-98	40	56	45	30	–	387	1564
AU-SA-96	60	43	36	20	–	296	1876
AU-TE-99	30	37	26	13	–	308	806
BR-SA-00	25	14	–	6	–	63	150
BR-SM-00	25	23	–	12	–	127	300
BR-SN-00	25	30	–	14	–	140	350
DK-FG-12	50	90	69	–	279	1077	1077
DK-HG-12	50	100	71	–	523	1235	1235
DK-VG-09	60	46	53	–	163	918	918
ES-SS-08	35	66	4	21	–	225	439
FI-PB-98	40	46	34	31	–	387	854
FI-WP-06	35	18	13	10	–	172	297
FI-MP-06	35	25	25	14	–	280	306
GR-PA-08	35	19	–	12	–	262	262
IT-I4-96	36	61	–	38	–	748	1101
KS-PR-11	62	101	–	63	–	809	1912
NL-KP-03	38	75	41	18	453	1156	1203
NL-KP-05	37	78	42	26	498	1235	1272
NL-KP-09	38	93	53	48	–	1148	1274
UK-SP-06	25	68	67	67	–	1227	1227
US-WS-09	100	134	108	–	–	628	6354
ZA-LW-09	148	19	2	16	–	185	838
ZA-WD-09	42	40	–	30	–	278	1353

Lastly, there are many *distribution* constraints that are evaluated between individual pairs of classes, or all classes as a whole. Distribution constraints may affect the time of the day, the days of the week, the week of the semester, or the room assigned (see Müller et al., 2018, §3.5).

A penalty is associated to the selection of a room and a period for a class, so that the objective function is composed by four main components: (*i*) class/period penalization, (*ii*) class/room penalization, (*iii*) violations of distribution constraints, and (*iv*) student conflicts.

Benchmarks. Instances have been collected using the non-commercial timetabling system UniTime (https://www.unitime.org), and they come from ten institutions, including Purdue University (USA), Masaryk University (Czech Republic), AGH University of Science and Technology (Poland), and Istanbul Kültuür University (Turkey). The real-life data was properly anonymized and simplified as discussed below.

The dataset is composed by 30 instances from ITC-2019 (10 early, 10 middle, 10 late) with very different features in terms of size of the instance (number of classes, students and rooms), room utilization, student course demand, course structure, time patterns, travel times and distribution constraints. Such diversity reflects the different sources of the data, both for the type of institution (school/faculty/entire university) and geographical position. Table 7 reports a selection of the instance features available from a more comprehensive list published on the competition website. The symbol – indicates that the corresponding feature cannot be computed because the number of students is null for that instance.

File formats and repositories. Instances are written in XML format and available from the competition website (https://www.itc2019.org) after registering. In addition, the winners of ITC-2019 have implemented a preprocessing procedure for the ITC-2019 dataset (Holm, Mikkelsen, Sørensen, & Stidsen, 2020) that reduces instances to a simplified, though still complete, form. The reduced ITC-2019 dataset is available at https://dsumsoftware.com/itc2019/.

Other datasets and generators. Up to our knowledge, there are no other instances available apart from the six test instances provided in the competition website.

Discussion. As mentioned above, this formulation thoroughly adheres to real-world situations, and can be consider a big step ahead in the process of bringing practical timetabling to be solved by academic research methods. Indeed, it considers many real-life features that have been neglected in the previous formulations. In particular, the integration of course timetabling and student sectioning is an important novelty. In addition, the possibility to have different timetables in the various weeks of the semester is another feature that many university nowadays use in common practice. Nonetheless, a few aspects of real-life data have been still neglected or transformed into existing constraints in order to make the formulation easier to model and to work on.

One simplification regards the usage of rooms. In some cases, a penalty occurs when the room is too big for the current number of students; other cases involves a complex sharing policy of rooms between departments. Some other real-world situations consider a finer tuning of the travel times between rooms, based on precise distances. Lastly, in some cases there is the possibility that a student makes a reservation, thus gaining priority to be assigned to a specific class. These simplifications however regard a limited set of cases, and do not reduce significantly the generality of the formulation.

3.7. Other formulations

We now review the additional problem formulations that provide real-world instances that are publicly available. Table 8 shows the list of available ones, up to our knowledge, along with the information whether the solutions and a solution checker are available.

There are many papers claiming that the search method has been applied to real-world cases, but then they do not provide the corresponding files (mainly for privacy issues). There are also many cases in which the link for retrieving the instances is not working anymore, typically due to authors changing affiliation. The latter phenomenon clearly show that the strategy of posting data in author's website does not work in the long run. In some cases, the link has been restored by the authors upon our specific request.

As mentioned in the introduction, a trustworthy method to assess the quality of a new search method for a novel problem is through solving some of the benchmark instances, suitably adapted, with this new method. This approach has been taken by Phillips, Waterer, Ehrgott, & Ryan (2015) that solve a novel CTT problem, but also the CB-CTT benchmarks, and by Woumans, De Boeck, Beliën, & Creemers (2016) that solve their ETT problem,

Table 7
Features of the ITC-2019 benchmark instances. Symbol definition: **Co** (Courses), **Cl** (Classes), **R** (Rooms), **S** (Students), **W** (Weeks), **CoS** (Courses per Student), **ClS** (Classes per Student), **MM** (Minutes per Meeting), **RCl** (Rooms per Class).

Inst.	Co	Cl	R	S	W	CoS	ClS	MM	RCl
agh-fis-spr17	340	1239	80	1641	16	8.17	16.2	104.46	15.92
agh-ggis-spr17	272	1852	44	2116	16	6.98	29.92	124.46	7.28
bet-fal17	353	983	62	3018	16	6.24	9.08	93.38	25.43
iku-fal17	1206	2641	214	0	14	–	–	123.7	30.76
mary-spr17	544	882	90	3666	16	2.88	2.9	141.88	13.57
muni-fi-spr16	228	575	35	1543	15	6.24	10.06	121.45	4.82
muni-fsps-spr17	226	561	44	865	19	7.76	11.6	90.97	3.15
muni-pdf-spr16c	1089	2526	70	2938	13	8.72	17.35	140.5	11.82
pu-llr-spr17	697	1001	75	27,018	16	3.03	3.4	63.52	15.23
tg-fal17	36	711	15	0	14	–	–	132.53	4.41
agh-ggos-spr17	406	1144	84	2254	16	7.01	13.94	93.51	10.92
agh-h-spr17	234	460	39	1988	16	2.6	4.18	113.71	25.47
lums-spr18	313	487	73	0	20	–	–	101.46	27.19
muni-fi-spr17	186	516	35	1469	14	6.22	10.3	124.55	5.25
muni-fsps-spr17c	116	650	29	395	14	6.98	32.94	110.88	5.06
muni-pdf-spr16	881	1515	83	3443	13	9.2	10.04	84.93	17.47
nbi-spr18	404	782	67	2293	15	6.03	12.46	106.13	4.83
pu-d5-spr17	212	1061	84	13,497	15	1.45	2.46	86.73	8.77
pu-proj-fal19	2839	8813	768	38,437	17	4.71	6.95	87.37	9.83
yach-fal17	91	417	28	821	16	5.07	13.14	113.21	4.61
agh-fal17	1363	5081	327	6925	18	8.7	20.91	106.29	10.52
bet-spr18	357	1083	63	2921	16	6.52	10.46	88.74	25.15
iku-fal18	1290	2782	208	0	13	–	–	127.15	27.72
lums-fal17	328	502	73	0	20	–	–	108.51	26.54
mary-fal18	540	951	93	5051	16	4.16	4.17	138.57	15.11
muni-fi-fal17	188	535	36	1685	13	6.59	10.43	122.54	4.94
muni-fspsx-fal17	515	1623	33	1152	21	8.87	21.82	111.82	4.42
muni-pdfx-fal17	1635	3717	86	5651	13	9.84	15.94	131.45	18.48
pu-d9-fal19	1154	2798	224	35,213	15	3.51	4.37	74.98	14.24
tg-spr18	44	676	18	0	16	–	–	137.9	5.67

Table 8
Other formulations and datasets. #Inst: number of instances, Sol: solutions available (√ = Yes, × = No), Check: checker available, Format: file format, Source: single institution or country in case of many institutions, URL: web link.

Reference	Prob	#Inst	Sol	Check	Format	Source	URL
Beligiannis, Moschopoulos, Kaperonis, & Likothanassis (2008)	HTT	11	×	×	text	Greece	https://www.dropbox.com/s/rolhmd31bmrea4a/Input%20instances.zip
Rudová, Müller, & Murray (2011)	CTT	50	√	√	XML	Purdue (US)	https://www.unitime.org
Müller (2016)	ETT	9	√	√	XML	Purdue (US)	https://www.unitime.org
Woumans et al. (2016)	ETT	1	√	×	Excel	Belgium	https://www.kuleuven.be/cv/personallinks/u0038694e.htm
Saviniec & Constantino (2017)	HTT	34	×	×	XML	Brazil	https://www.gpea.uem.br/benchmark.html
Lemos, Melo, Monteiro, & Lynce (2019)	CTT	8	√	√	XML	Lisbon (PT)	https://github.com/ADDALemos/MPPTimetables
Battistutta et al. (2020)	ETT	40	√	√	JSON	Italy	https://bitbucket.org/satt/examtimetablinguniuddata
Güler, Geçici, Köroğlu, & Becit (2021)	CTT	1	×	×	Excel	Yıldız (TR)	https://sites.google.com/view/mgguler/datasets

but discuss also the application to two UETT instances (sta83 and yor83).

4. State-of-the-art results

In this section, we report the results for each of the formulations introduced in Sections 3.1–3.6. For each one, among all results in the literature, we select and report the ones that we consider "state-of-the-art", intending with this term those that have the best scores for some instances. However, in this selection we take into account also the running time, thus including also results that are worse than others but obtained with significantly shorter time.

For each contribution, we show, if available, the average and the best scores for each instance, along with the running time (when relevant). Further details, such as the computing speed and the number of threads of the machines are neglected, and can be retrieved (if reported) in the corresponding articles.

The tables include also, when available, the best lower bound and the best known result (upper bound), specifying also the researchers that have found them. In addition, the lowest best values are in italics and the proven optimal solutions are underlined (except for perfect solutions). Finally, top average results in the table are in boldface.

4.1. Results on UETT

The methods proposed for UETT in the literature have different running times. Therefore a completely fair comparison is not possible, given that UETT is particularly sensible to the running time. In fact, longer runs consistently produce results better than shorter ones. As a consequence, in order to select which are the state-of-

the-art results, we consider both the scores and the running time in sort of a two-objective view.

The selected state-of-the-art results on Toronto benchmarks described in Section 3.1 are shown in Table 9. The last two columns report the LBs and the best UBs. The UBs are obtained by several authors, whereas the LBs are obtained by Gogos, Dimitsas, Nastos, & Valouxis (2021) and Dimitsas, Nastos, Valouxis, Alefragis, & Gogos (2022) by using *ad hoc* procedures. The letter beside each LB and UB value indicates who are the authors: "G" stands for Gogos et al. (2021), "D" for Dimitsas et al. (2022), "BB" for Burke & Bykov (2016), "L" for Leite, Fernandes, Melício, & Rosa (2018) and "BC" for Bellio et al. (2021). For sta83 the UB has been reached by several authors, and we marked it with the symbol $*$.

As mentioned above, the proposed methods have different running times, which are reported in the right-most three columns of Table 9, so that highlighted values do not identify univocally the "best" contributions, as they compete with different timeouts. For this reason, for Bellio et al. (2021) we report the results of both the short and long runs, even though the short ones are clearly inferior to the long ones, but can be considered as competitive for the allotted time.

We notice that all methods used to find the UBs are metaheuristics, and in particular they are based on local search. They range from Simulated Annealing (Bellio et al., 2021), to various versions of Hill Climbing and Great Deluge (Burke & Bykov, 2016), to Memetic Algorithms (Leite et al., 2018), which are hybrid techniques using also local search. Indeed, there are no approaches such as mathematical and constraint programming among the most successful ones. As we will see in the next sections, this is not the case for some of the other formulations (see Sections 4.3, 4.5, and 4.6).

We can see that, unfortunately, the LBs by (Gogos et al., 2021) are not particularly tight, leaving room for improvements. On the contrary, the single LB on sta83 by Dimitsas et al. (2022) is tight, and actually proves the optimality of the best cost 157.03 found by several authors

We remark that there are some early results for which it is not clear whether they were obtained on the original input data (see discussion on Section 3.1). Therefore, we decided to remove them and to bound to fully trustworthy results only.

4.2. Results on PE-CTT

For the PE-CTT formulation, the results that we consider state-of-the-art are shown in Tables 10, 11 and 12, for the two datasets identified as benchmarks in Section 3. The second dataset is split into two tables because the first set of 16 instances and the second one of 8 instances have been considered by different authors, as the latter have been released at a later stage.

All results are obtained from 31 runs, using the time limit allowed by the competition benchmark program (about 300s). For each instance, the top average result is shown in boldface, whereas the lowest best value is shown in italic. For the ITC-2002 benchmarks, the column UB of Table 10 reports the best known value, which in this case is the lowest value in the table, except for instance 1 for which it has been obtained by Goh et al. (2017) with longer (five times) timeout, and instances 10 and 11 obtained by Nagata (2018) using a method different from the most performing one reported here. The first column reports the contribution by Kostuch (2005), who was the winner of the ITC-2002 competition.

We do not report the UBs in Tables 11 and 12 as most of them are equal to 0 (for instance 11, UB is also 0, as found by Lewis & Thompson 2015, not reported in Table 11). The only distinctive instances are 3, 4, and 20 with UB values 55, 10 (reported in the corresponding table), and 150 (found by Nagata 2018, with another method), respectively. For ITC-2002, conversely, for many in-

Table 9
Results on Toronto benchmarks of UETT.

Inst.	Bellio et al. 2021		Burke & Bykov 2008		Mandal, Kahar, & Kendall 2020		Burke & Bykov 2016		Leite et al. 2018		Bellio et al. 2021		LB	UB
	avg	best	avg	best	avg	best	avg	best	avg	best	avg	best		
car91	4.44	4.38	4.68	4.58	4.72	4.58	4.34	4.32	4.39	4.31	**4.27**	*4.24*	0.0059^{G}	4.237932^{BC}
car92	3.8	3.75	3.92	3.81	3.93	3.82	3.7	3.67	3.72	3.68	**3.68**	*3.64*	0.0079^{G}	3.642109^{BC}
ear83	32.89	32.61	32.91	32.65	34.49	33.23	32.66	32.62	32.61	32.48	**32.60**	*32.42*	18.2596^{G}	32.420444^{BC}
hec92	10.16	10.05	10.22	10.06	11.09	10.32	10.12	10.06	**10.05**	*10.03*	**10.05**	*10.03*	3.8162^{G}	10.033652^{L,BC}
kfu93	13.06	12.87	13.02	12.81	13.97	13.34	12.85	*12.8*	**12.83**	*12.81*	12.88	*12.81*	5.736^{G}	12.799028^{BC}
lse91	10.09	9.92	10.14	9.86	10.62	10.24	9.84	9.78	9.81	9.78	**9.80**	9.78	3.3555^{G}	9.773661^{BC}
pur93	4.32	4.22	4.71	4.53			**3.91**	3.88	4.18	4.14	4.02	4	0.0014^{G}	3.88^{BB}
rye93	8.1	7.99	8.06	7.93	10.29	9.79	7.94	7.91	7.93	7.89	**7.91**	*7.84*	3.7868^{G}	7.837586^{BC}
sta83	157.05	*157.03*	157.05	*157.03*	157.64	157.14	157.04	*157.03*	**157.03**	*157.03*	**157.03**	*157.03*	157.03^{D}	157.03*
tre92	7.85	7.72	7.89	7.72	8.03	7.74	7.68	7.64	7.7	7.66	**7.66**	7.59	0.8601^{G}	7.590367^{BC}
uta92	3.13	3.05	3.26	3.16	3.22	3.13	3.01	2.98	3.04	3.01	**2.97**	2.95	0.0022^{G}	2.947193^{BC}
ute92	24.82	24.76	24.82	24.79	26.04	25.28	24.82	24.78	24.83	24.8	**24.79**	24.76	21.5993^{G}	24.76^{BC}
yor83	34.93	34.56	36.16	34.78	36.79	35.68	34.79	34.71	34.63	34.45	**34.57**	34.40	19.1435^{G}	34.404888^{BC}
Time														
Min	130.8	s	450.0	s			4.6	h	24	h	26.2	h		
Max	1382.0	s	901.0	s			5.7	h	48	h	52.2	h		
Avg	413.1	s	654.6	s	1	h	5.1	h	31.4	h	34.7	h		

Table 10
Results on ITC-2002 benchmarks of PE-CTT.

	Kostuch	Goh et al.		Nagata		Goh et al.		Goh et al.		
	2005	2017		2018		2019		2020		
Inst.	best	avg	best	avg	best	avg	best	avg	best	UB
01	*16*	32.6	23	**30.2**	*16*	37	26	36.8	29	10
02	*2*	13.7	7	**11.4**	*2*	16.3	6	16.2	*2*	2
03	*17*	36.4	26	**31**	*17*	38.2	27	34.3	24	17
04	*34*	63.1	50	**60.8**	*34*	69	47	70.7	46	34
05	42	58.6	38	72.1	42	**51.8**	*36*	55	43	36
06	0	0.8	0	2.4	0	0.8	0	**0.4**	0	0
07	2	2.6	0	8.9	2	**2.4**	0	**2.4**	0	0
08	0	**1.4**	0	2	0	1.5	0	2.2	0	0
09	1	**4.6**	0	5.8	2	6.4	0	6.5	0	0
10	*21*	40.9	28	**35**	*21*	40.4	22	39.2	26	18
11	*5*	17.7	10	**12.9**	*5*	19	10	19.7	9	4
12	55	64.5	53	76.3	55	64.1	47	**63.9**	*46*	46
13	*31*	53.3	38	**47.1**	*31*	51	33	51.2	40	31
14	11	12.9	5	22.3	11	13.6	*4*	**12.1**	*4*	4
15	2	**4.0**	0	8.4	2	4.8	0	4.4	0	0
16	0	**0.5**	0	3.4	0	2.2	0	1.6	0	0
17	37	41.6	26	54	37	**36.8**	25	38.7	*24*	24
18	4	9.7	*2*	**9.4**	4	12.5	3	11.7	4	2
19	*7*	24.7	11	**16.4**	*7*	25.6	15	23.6	9	7
20	0	**0**	0	0.5	0	**0**	0	**0**	0	0
Avg		24.18		25.52		24.67		24.53		

Table 11
Results on ITC-2007 public benchmarks of PE-CTT.

	Mayer, Nothegger, Chwatal, & Raidl		Cambazard et al.		Goh et al.		Nagata		Goh et al.		Goh et al.	
	2008		2012		2017		2018		2019		2020	
Inst.	avg	best	avg	best	avg	best	avg	best	avg	best	avg	best
1	613	0	547	15	307.6	0	**81.7**	0	209.4	0	191.8	0
2	556	0	403	356	63.4	0	48	0	10.1	0	**1.7**	0
3	680	110	254	174	199.4	163	**155**	*55*	188.6	141	189.8	137
4	580	53	361	249	328.8	242	**254.1**	*10*	320.9	192	315.5	24
5	92	13	26	0	2.7	0	**0**	0	2.9	0	2.9	0
6	212	0	16	0	33.2	0	**0**	0	54.7	0	37.6	0
7	4	0	8	1	18	5	**3.6**	0	14.5	4	16.2	5
8	61	0	**0**	0	**0**	0	**0**	0	1.6	0	5.7	0
9	202	0	1167	29	100.7	0	**58.9**	0	15.2	0	2.6	0
10	**4**	0	*1297	2	65.3	0	6.4	0	30.5	0	16.3	0
11	774	143	361	178	244.3	161	**140.4**	*3*	201.6	136	199.6	21
12	538	0	380	14	318.2	0	**33.1**	0	303.5	0	258.1	0
13	360	5	135	0	99.5	0	**0**	0	90.4	0	85.9	0
14	41	0	15	0	0.2	0	**0**	0	25.6	0	17.8	0
15	29	0	47	0	192	0	**0**	0	12.5	0	9.3	0
16	101	0	58	1	105.8	10	**1.5**	0	45.8	0	40.2	0
Avg	302.9		317.2		129.9		48.9		95.5		86.9	

Table 12
Results on ITC-2007 hidden benchmarks of PE-CTT.

	Cambazard et al.		Ceschia et al.		Lewis and Thompson		Goh et al.		Nagata		Goh et al.		Goh et al.	
	2012		2012		2015		2017		2018		2019		2020	
Inst.	avg	best	avg	best	avg	best	avg	best	avg	best	avg	best	avg	best
17	4.9	0	**0.0**	0	0.07	0	0.8	0	0.2	0	0.5	0	0.1	0
18	14.1	0	41.1	0	2.16	0	12.5	0	**0.5**	0	7.7	0	15.5	0
19	2027.0	1824	951.5	0	346.08	0	516.7	0	616.8	0	**11**	0	79.6	0
20	505.0	445	700.2	543	724.54	557	650.7	586	**482**	*438*	664	555	661.5	579
21	27.1	0	35.9	5	32.09	1	12.5	0	**0.1**	0	25.7	0	14.8	0
22	550.8	29	19.9	5	1790.08	4	136	1	35	0	**5.8**	0	22.6	0
23	**330.5**	238	1707.7	1292	514.13	0	504.4	11	1083.5	777	713.6	56	531.7	0
24	124.2	21	105.3	0	328.18	18	192.6	5	1	0	**77.5**	0	102.1	0
Avg	448.0		445.2		467.2		253.3		277.4		188.2		178.5	

Table 13
Results on ITC-2007 benchmarks of CB-CTT.

	Abdullah & Turabieh 2012		Kiefer et al. 2017		Lindahl et al. 2018		
	avg	best	avg	best	avg	LB	UB
comp01	**5.00**	*5*	**5.0**	*5*	12.0	$5^{C,B1}$	5^{*}
comp02	**36.36**	*26*	41.5	34	49.5	24^{B2}	24^{A}
comp03	74.36	70	**71.7**	*68*	74.5	58^{B3}	64^{K}
comp04	38.45	*35*	**35.1**	*35*	38.5	35^{*}	35^{*}
comp05	314.45	295	**305.2**	*294*	373.5	247^{B3}	284^{K}
comp06	**45.27**	*30*	47.8	41	58.3	27^{A}	27^{A}
comp07	**12.00**	*7*	14.5	10	35.0	6^{*}	6^{A}
comp08	**40.82**	*37*	41.0	39	49.7	37^{*}	37^{A}
comp09	108.36	102	102.8	*100*	**100.5**	96^{B2}	96^{L1}
comp10	**8.36**	*5*	14.3	7	25.7	4^{*}	4^{A}
comp11	0.00	*0*	0.0	*0*	6.5	0^{*}	0^{*}
comp12	320.27	315	**319.4**	*306*	360.7	248^{B3}	294^{K}
comp13	64.27	*59*	**60.7**	*59*	69.0	59^{*}	59^{A}
comp14	64.36	61	**54.1**	*51*	56.9	51^{*}	$51^{A,L1}$
comp15	72.73	69	**72.1**	*66*	74.5	58^{B3}	62^{K}
comp16	**23.73**	*18*	33.8	26	37.1	$18^{A,B2}$	18^{A}
comp17	76.36	*60*	**75.7**	67	86.1	$56^{A,B3}$	56^{A}
comp18	75.64	69	**66.9**	*64*	72.9	61^{L2}	61^{K}
comp19	66.82	*57*	**62.6**	59	64.8	57^{B2}	57^{L1}
comp20	**13.45**	*7*	27.2	19	34.3	4^{*}	4^{A}
comp21	100.73	*86*	**97.0**	93	103.8	$74^{B2,L2}$	74^{P}
Avg	74.37	67.29	**73.73**	68.71	84.94		

stances, the perfect solution is still to be found. For Cambazard, Hebrard, O'Sullivan, & Papadopoulos (2012), who won the Track 2 of ITC-2007, the $*$ symbol indicates that not all solutions are feasible and the average is computed only on the feasible ones. The percentage of feasible solutions for instance 10 is 89%, whereas in all other cases is 100%.

The first comment on these tables is that all best results have been found by local search methods, namely Tabu Search (Nagata, 2018) and Simulated Annealing (Goh et al., 2017; 2019; Goh, Kendall, Sabar, & Abdullah, 2020). In general, best results are obtained by Nagata (2018), that uses a composite neighborhood and *elite candidate* rules to reduce the computational cost of the full neighborhood exploration prescribed by Tabu Search. Good results are obtained also by Goh et al. (2017, 2019); Goh et al. (2020), mainly using random move selection.

Goh, Kendall, & Sabar (2019) report also the results for longer running times (i.e., five times longer), which are not shown here. Unsurprisingly, both the best and average cost are remarkably improved when the execution time is extended.

The fact that all instances have a perfect (zero cost) solution might bias the search methods toward certain specific strategies. For example, the objective that penalizes all lectures in the last period of the day might be exploited, by removing such periods completely from the search space.

4.3. Results on CB-CTT

Table 13 shows the best results for CB-CTT benchmarks obtained using the timeout fixed for the ITC-2007 dataset (300–500 s depending on the CPU). Longer runs, which unsurprisingly obtain better results, are not considered here (see Asín Achá & Nieuwenhuis, 2014; Lü & Hao, 2009; Song et al., 2021). We take them into account only for establishing the LBs and UBs, which are shown in the last two columns of the table. In particular, the LBs are obtained with a running time up to 40 times the ITC-2007 timeout.

Besides each best-known lower and upper bound values, we report a letter that indicates who are the authors[3]: "A" stands for Asín Achá & Nieuwenhuis (2014), "B1" for Burke, Mareček, Parkes, & Rudová (2010), "B2" for Bagger, Desaulniers, & Desrosiers (2019b), "B3" for Bagger, Sørensen, & Stidsen (2019a), "C" for Cacchiani, Caprara, Roberti, & Toth (2013), "K" for Kiefer, Hartl, & Schnell (2017), "L1" for Lü & Hao (2009), "L2" for Gerard Lach, "P" for Phillips (2015). If the same value was found by many different authors, we marked it with the symbol $*$.

We see from Table 13 that almost all current best results are obtained by two contributions, namely Abdullah & Turabieh (2012) and Kiefer et al. (2017), who both proposed metaheuristic methods using Adaptive Large Neighborhood operators. Abdullah & Turabieh implemented a Genetic Algorithm hybridized with Tabu Search employing large neighborhood operators, whose sequence of employment follows a "best" selection strategy, based on previous knowledge about the successful percentage of each neighborhood structure on each instance. Kiefer et al. (2017) presented an Adaptive Large Neighborhood Search algorithm embedded in a Simulated Annealing framework, incorporating several destroy and repair operators, whose selection probability is dynamically biased towards the best-performing ones. Several other papers have produced results that were state-of-the-art at the time of their publication, including (Abdullah, Turabieh, McCollum, & McMullan, 2012; Bellio, Ceschia, Di Gaspero, Schaerf, & Urli, 2016; Lü & Hao, 2009; Müller, 2008). In particular, the solver by Müller (2008) won the track of ITC-2007 dedicated to CB-CTT.

As remarked by Bagger et al. (2019a, Table 6), all benchmark instances but 3 are currently solved to optimality[4] In our opinion, the fact that the optimal value has been found does not undermine the benchmarking role of these instances, which are still challenging for medium-short timeouts. Nonetheless, there are other public instances that are already available (on OptHub) that could come up beside the current ones, in order to create a larger, more comprehensive benchmark set (see Section 3.3).

[3] We note that the UBs and LBs in Table 4 of Lindahl, Sørensen, & Stidsen (2018) (column Best, including the numbers in parentheses) are actually wrong, as they refer to the formulation UD1 instead of UD2 considered in that paper (and here).

[4] In the paper they are actually 4, but as mentioned above, `comp03` and `comp15` are identical in this formulation.

Table 14
Results on ITC-2007 benchmarks of ITC-2007-ETT.

	Burke & Bykov 2016		Bykov & Petrovic 2016	Burke & Bykov 2017	Gogos, Goulas, Alefragis, Kolonias, & Housos 2010	Arbaoui et al. 2019
Inst.	avg	best	best	avg	best	LB
1	3792.5	3691	*3647*	**3787**	4128	–
2	**393.1**	385	385	402	*380*	10
3	7611.8	*7359*	7487	**7378**	7769	670
4	**12100.4**	*11329*	11,779	13,278	13,103	1620
5	2512.9	2482	*2447*	**2491**	2513	–
6	25491.5	25,265	*25210*	**25,461**	25,330	22875*
7	3755.1	3608	*3563*	**3589**	*3537*	–
8	6949.9	6818	*6614*	**6701**	7087	1250*
9	**930**	*902*	924	997	913	–
10	**12975.7**	*12900*	12,931	13,013	13,053	0
11	23931.7	*22875*	23,784	**22,959**	24,369	3970
12	**5176.3**	5107	5097	5234	*5095*	2030

It is worth noticing that this is the only one among our six standard formulations for which there has been a lot of research improving the lower bounds.

4.4. Results on ITC-2007-ETT

The state-of-the-art results for ITC-2007-ETT using the ITC-2007 timeout are shown in Table 14. First of all, we notice that the best results are obtained mainly by Bykov and coworkers. They use innovative local search algorithms, such as Late Acceptance and Step Counting Hill Climbing, applied to complex neighborhood structures (such as Kempe chains). The corresponding track of the ITC-2007 competition was won by Müller (2008), whose results however are not currently competitive with the current bests shown in the table.

Research on this problem is still active and more recent results are available (see, e.g., Battistutta et al., 2017; Leite, Melício, & Rosa, 2019; 2021); however, they do not outperform the previous ones shown in Table 14.

The lower bounds are obtained by Arbaoui, Boufflet, & Moukrim (2019) by considering only a subset of the soft constraints. In detail, they consider the spacing soft constraints, namely (*i*) and (*ii*) mentioned in Section 3.4, and compute the number of violations induced by the largest clique in the corresponding graph. As shown in Table 14, for some instances the method does not produce any result as the largest clique is not big enough to contribute any violation. For instances 6 and 8, marked with an *, we add to the LB computed by Arbaoui et al. (whose original values were 2600 and 0, respectively) the fixed cost of constraints (*iii*) and (*v*) due to the fact that there are not enough periods to satisfy them (see Section 3.4 for the detailed explanation).

4.5. Results on XHSTT

Table 15 reports the state-of-the-art results for the benchmark instances of XHSTT. As mentioned in Section 3.5, two instances have been eliminated due to the fact that they are too easy.

The first three columns report average results obtained within the competition timeout (1000 secs), whereas Fonseca, Santos, Carrano, & Stidsen and Kheiri & Keedwell did not impose a time limit. The column *z* reports the output of one single deterministic run. Besides each best-known lower and upper bound values, we report a letter that indicates who are the authors: “G” stands for the UFOP-GOAL team (Fonseca, Santos, & Carrano), “L” for the Lectio team (Kristiansen, Sørensen, & Stidsen), “V” for the VAGO team (Valouxis, Gogos, Daskalaki, Alefragis, Goulas, and Housos), “D” for Á. P. Dorneles, “V2” for M. de Vos, “D2” for Demirović & Musliu, and “S” for Skolaris (M. Klemsa).

We first notice that most authors do not report the results for all instances. There are various reasons for this behavior. For example, Fonseca et al. (2017) omit instances whose optimal solution was already known and proven. Furthermore, the method by Demirović & Stuckey (2018) does not support all possibles constraints, so that they excluded some instances. Some others have considered previous versions of the archive, that contained different instances. Finally, for some instances the method by Demirović & Musliu (2017) did not return any solution within the timeout granted or it incurred in some other problems (such as memory exceeded).

Although the ITC-2011 competition was dominated by metaheuristic methods, and in particular won by Fonseca, Santos, Toffolo, Brito, & Souza (2016b), recently exact methods based on integer programming (Dorneles, de Araújo, & Buriol, 2017; Fonseca et al., 2017; Kristiansen et al., 2015), maxSAT (Demirović & Musliu, 2017) and constraint programming (Demirović & Stuckey, 2018) have proven to be very effective for XHSTT. Differently from the formulations of Sections 3.1–3.4, for XHSTT it ended up being customary to use IP techniques and to evaluate the performance of a solution methods without time limit. Indeed, its best known solutions and lower bounds are updated/improved by the community on the XHSTT website. An up-to-date categorization of the different solution methods applied to HTT (including XHSTT) is presented by Tan et al. (2021).

As mentioned in Section 3.5, the previous versions of the XHSTT archive are deprecated, and thus we do not include them in the benchmarks. However, one of them, namely the hidden dataset of ITC-2011, due to its popularity given by the competition, has been used as testbed by many authors. In particular, there are interesting results by Kristiansen et al. (2015), Fonseca, Santos, & Carrano (2016a), Demirović & Musliu (2017), and Teixeira, Souza, de Souza, & Coelho (2018). In addition, LBs have been found by Dorneles et al. (2017).

4.6. Results on ITC-2019

The competition finished in 2020, so the problem is rather new, and the only published results are those of the competitors. Differently from previous competitions, the goal of ITC-2019 was to find all-time-best solutions to all competition instances, without time limits or technology restrictions. In the last column of Table 16 we report the UBs whose solutions are collected (and continuously updated) on the competition website (https://www.itc2019.org/).

The competition was won by the DSUM team (Holm et al., 2020) who devised a Fix-and-Optimize matheuristic, which was able to find all best solutions except for one instance (agh-fal17). In addition, the DSUM team maintains a website (https:

Table 15
Results on the XHSTT-2014 benchmarks of XHSTT.

Inst.	Demirović and Musliu 2017 avg	Demirović and Stuckey 2018 avg	Teixeira et al. 2018 avg	Fonseca et al. 2017 z	Kheiri and Keedwell 2017 best	LB	UB
AU-BG-98			(3, 514)	129	493	0	128^{G}
AU-SA-96			(16, 91)	*0*	2	0	0^{G}
AU-TE-99			(7, 13)	*20*	61	20^{G}	20^{G}
BR-SA-00	5	5			10	$5^{L,D}$	5^{L}
BR-SM-00	61.4	88	100		(2, 117)	$51^{L,D}$	51^{L}
BR-SN-00	50.6	66	170		101	35^{D}	35^{D}
DK-FG-12				1300	1522	412^{G}	1263^{G}
DK-HG-12				(12, 2356)	(12, 2628)	7^{G}	$(12, 2330)^{G}$
DK-VG-09				(2, 2329)	(2, 2720)	$(2, 0)^{G}$	$(2, 2323)^{G}$
ES-SS-08				*335*	517	$334^{L,G}$	335^{L}
FI-PB-98	54.6	9			8	0	0
FI-WP-06	9.8	4	1		7	0	0^{G}
FI-MP-06	95.2	90	93		89	$77^{L,V2}$	77^{G}
GR-PA-08	5	7			4	3^{L}	3^{G}
IT-I4-96	35				34	27^{L}	27^{G}
KS-PR-11					3	0	0^{D2}
NL-KP-03			1383	*199*	466	0	199^{G}
NL-KP-05			1056	433	811	$89^{V2,G}$	425^{G}
NL-KP-09				*1620*	(2, 7495)	180^{G}	1620^{G}
UK-SP-06				(5, 4014)	(19, 1294)	0	$(4, 1708)^{S}$
US-WS-09				103	512	101^{G}	101^{G}
ZA-LW-09		*0*	28		52	0	0^{V}
ZA-WD-09	1.2				(9, 0)	0	0^{L}

Table 16
Results on ITC-2019 benchmarks of ITC-2019.

Inst.	Rappos et al. 2022 best	Holm et al. 2022 best	Mikkelsen & Holm 2022 best	DSUM TEAM LB	UB
agh-fis-spr17	4557	–	3463	1411	2985^{D}
agh-ggis-spr17	36,616	–	38,026	23,164	$34,285^{D}$
bet-fal17	295,427	–	319,059	89,278	$289,656^{D}$
iku-fal17	26,840	–	19,498	18,099	$18,968^{D}$
mary-spr17	15,021	15,174	14,924	14,472	14910^{D}
muni-fi-spr16	3844	7741	3766	3621	3756^{D}
muni-fsps-spr17	883	868	868	868	868^{D}
muni-pdf-spr16c	37,487	–	66,812	16,255	33331^{D}
pu-llr-spr17	13,385	10,107	10,055	10,038	10038^{D}
tg-fal17	4215	4215	4215	4215	4215^{U}
agh-ggos-spr17	6320	–	4652	1982	2855^{D}
agh-h-spr17	26,159	–	23,883	8945	21161^{D}
lums-spr18	114	95	95	24	95^{D}
muni-fi-spr17	4289	10,093	3845	2512	3738^{D}
muni-fsps-spr17c	3303	509,503	3777	1361	2594^{D}
muni-pdf-spr16	24,318	–	22,533	13,626	$17,159^{D}$
nbi-spr18	19,055	18,014	18,014	18,014	18014^{D}
pu-d5-spr17	18,813	–	17,731	6981	$15,185^{M}$
pu-proj-fal19	561,194	–	219,832	67,549	$117,186^{M}$
yach-fal17	1844	–	1717	526	1074^{M}
agh-fal17	–	19,046	261,826	6522	117627^{M}
bet-spr18	360,057	–	375,677	76,489	$348,524^{D}$
iku-spr18	36,711	–	28,436	25,855	$25,863^{D}$
lums-fal17	386	369	349	254	349^{D}
mary-fal18	5637	–	4546	3546	4331^{D}
muni-fi-fal17	3794	9177	3199	1890	2837^{D}
muni-fspsx-fal17	33,001	–	36,461	7869	10058^{M}
muni-pdfx-fal17	151,464	–	138,916	29,333	$82,258^{D}$
pu-d9-fal19	134,009	–	47,938	29,903	$39,081^{D}$

//dsumsoftware.com/itc2019/) reporting their current best results and the lower bounds (showed in Table 16). The second place was obtained by Rappos, Thiémard, Robert, & Hêche (2022) who modeled the problem as MIP enhanced with some preprocessing techniques that improve its efficiency. Table 16 reports their results obtained without time limits. The third place was occupied by Gashi, Sylejmani, & Imery (2021) who presented a Simulated Annealing algorithm[5]. In subsequent works, the DSUM team extended the graph-based reformulation used during ITC-2019 (Holm, Mikkelsen, Sørensen, & Stidsen, 2022) and conceived its parallelized version

[5] Their source code is available at https://github.com/edongashi/itc-2019

with solution sharing between threads (Mikkelsen & Holm, 2022). Table 16 shows their results with a time limit of 24 h. The symbol "–" denotes that no solution has been found.

The letter beside each UB value in Table 16 indicate the authors: "D" stands for the DSUM team (Holm, Mikkelsen, Sørensen, & Stidsen), "U" for the UFOP team (M, A. Pires, H. Gambini Santos, T. A.M. Toffolo), and "M" for Müller (2020).

5. Conclusions and future directions

The quest for formulations and benchmarks carried out for this survey has brought out various aspects of the current practice in timetabling research. We summarize here our observations, and we split them in three groups regarding the standard formulations, the specific formulations, and the solution techniques, respectively. In our opinion, these observations can serve as starting points for future research directions.

Key observations about standard formulations:

A. Most of the standard formulations arose from timetabling competitions, which have given the necessary initial boost in terms of infrastructure and promotion.
B. For some of the standard formulations, the benchmark instances are not challenging anymore, as they are too easily solved to optimality. Others, on the contrary, are still very challenging more than 20 years after their publication.
C. We can notice some common features across different formulations and benchmarks. First of all, the size of the instances in terms of events (lectures, exams,...) is rather uniform, with some exceptions, and around a few hundreds. This is due to the fact that they are mostly real-world cases, and this is the typical size of departments and other institutions. Secondly, the room occupancy is on average about 60%, which again is a reasonable value for balancing effectiveness and flexibility. On the contrary, we notice that the density of the conflict graphs are rather heterogeneous, ranging from 1% to more than 50%. Nonetheless, conflicts are not the only source of hardness, so that also instances with small conflict density are still challenging.
D. There is a clear trend in the timetabling community to move toward rich formulations, getting rid of strong simplifications. In our opinion, this is a positive trend, but should be paired with the maintenance and renewal of the simple formulations, that could still serve as better testbeds for comparisons.

Moving to the contributions introducing specific formulations, we have the following observations:

E. Many of the papers discussing original formulations do not provide publicly available data. For others, the original repository has become inaccessible after some time from the publication of the paper. Finally, in other cases, the file formats are too cumbersome and not sufficiently documented, to be easily usable for other researchers.
F. Most formulations are too specific for the particular case at hand without consideration of wider application, so that it is difficult to gain general insights from the papers. In addition, in some cases the precise formulation is not completely explained, so that it is not possible for other researchers to replicate the same model and to obtain comparable results.
G. For most formulations, the solutions are not made available, and thus the results in the papers could not be validated. In addition, the source code of the search method is very rarely available, so that the experiments cannot be replicated.
H. Only in a very few cases the solution of a specific formulation is complemented with the solution of some benchmarks of some standard formulation.

Despite the above "negative" observations, we nonetheless believe that these contributions still represent a praiseworthy effort for bridging the gap between theory and practice, by modeling and exposing to the community novel problems that take into account unsimplified real-life features.

Regarding the comparison of solution techniques, we make the following observations:

I. There is a need for the clear definition of the competition grounds, in terms of running time, statistical significance, computing architecture, usable technology, commercial licenses, and other issues. In the formulations coming form the competitions, the ground has been set by the official competition rules, which however might need to be refined and extended in order to do not harness future research.
J. The results of Section 4 clearly show that both exact and (meta)heuristic techniques have their role and their chance to emerge, depending on the specific formulation and the competition ground.
K. There is a need for new formulations and new benchmarks. In particular, for ETT the current benchmarks are still challenging, but there is a need for novel formulations that could better capture the real-world issues. For CTT, the ITC-2019 formulation is indisputably sufficiently close to real-world cases, but there might be room for alternative, possibly less complex, formulations. For HTT, in our opinion the main current concern is to collect new benchmarks that could take over for the ones that turned out to be too easy for state-of-the-art techniques.
L. There is also need for more instances that could be used for the statistically-principled tuning of the solution methods, letting the benchmarks to be used only for the validation phase (avoiding overtuning). To this aim, the use of high quality generators could also help, as these could provide an unlimited number of instances.

All above points together highlight the need for the development of research infrastructures in terms of common formulations, robust file formats, long-term web repositories with instances and solutions, generators, and solution checkers. The implementation of a wholesome and robust infrastructure of this type is clearly too expensive in terms of human effort to be left to the initiative of single research groups. Therefore, there is the need for coordinated community-level actions, in order to develop an infrastructure and, at the same time, create the necessary consensus upon its adoption. In our opinion, to this aim, the organization of future timetabling competitions could still be the right key to pursuit this task.

Another point that emerged from our analysis is the issue of the reproducibility and trustworthiness of results. In fact, the risk of reporting false results has emerged significantly, though mainly in the early times of the timetabling research. In any case, it is still important that data is available for both inspection and future comparisons. This is indeed a general issue that is ubiquitous in many research areas, as journals currently push for publication of data along with the papers.

We are trying to give our contribution for solving these issues by the development of the web application OptHub, which provides a common platform able to host new problems with their instances and solutions. Solutions in OptHub are immediately validated and made available to the community.

OptHub is an ongoing project, and hopefully will be extended significantly in future releases. The main future feature that will be include in a new version is the possibility to upload the software and to run it (also on behalf of other researchers). Hopefully, this option will allow the community to make fairer comparisons and statistical analyses on the behavior of the solution code.

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